



DIGESTION AND METABOLISM

DM IMCQ 4 ANAT

A 32-year-old man comes to the emergency department unresponsive after a motor vehicular collision. Vitals on admission were BP:85/60, RR:20x1, PR:110x1. Fluid resuscitation was started with two large bore needle catheters placed in the antecubital fossa of his upper limbs. Physical examination shows extensive abdominal bruising in the right upper quadrant. Ultrasound revealed “free” fluid (blood) in the peritoneal cavity. Follow up CT shows a liver laceration and abdominal surgery was planned. Which of the following peritoneal ligaments was most likely ligated during surgery to control the bleeding?

- A. Gastrocolic**
- B. Gastrophrenic**
- C. Hepatogastric**
- D. Gastrosplenic**
- E. Hepatoduodenal**

Describe the extent of the peritoneal cavity, the greater and lesser sacs, and the omental (epiploic) foramen (of Winslow).

Describe the contents of the hepatoduodenal ligament which will be gripped during a Pringle maneuver.

A 22-year-old woman comes to the emergency department because of fever, nausea, vomiting and right sided shoulder pain. Patient history reveals 14 days ago she was hospitalized for a pelvic abscess with accompanying peritonitis. She was treated with parenteral antibiotics and discharged from the hospital. At present her vitals are; BP: 110/80, PR:100x1. RR:20x1, Temperature:38.1°C (100.6°F). Physical examination shows right sided chest wall tenderness over the 8th to 11 ribs. There is also abdominal tenderness and decreased bowel sounds. CT scan shows an abscess located in the right upper quadrant of her abdomen. In which of the following locations will this abscess most likely be found?

- A. Hepatorenal recess**
- B. Subphrenic space**
- C. Paracolic gutter**
- D. Lesser sac**
- E. Pleural sac**

Describe the location of the referred pain from the gastrointestinal organs based on innervation and embryological origin.

Describe the location of the paracolic gutters, the vesicouterine pouch, the rectouterine pouch (of Douglas) and the hepatorenal pouch (of Morrison) within the peritoneal cavity.

A 4-day-old-baby boy comes to the intensive care unit because of high fever and refusal to feed. Physical examination shows abdominal distention and meconium coming from the external urethral opening. Normal perineal raphe with no anal opening is also noted . The results of his voiding cystourethrogram is shown in the image. Based on the clinical findings which of the following most likely explains the patient's condition?

- A. Urachal cysts
- B. Anal agenesis
- C. Anoperineal fistula
- D. Rectourethral fistula
- E. Omphaloenteric fistula



Describe the developmental mechanism of imperforate anus, anal agenesis, anal stenosis, membranous atresia of the anus, rectal agenesis, and anorectal agenesis

A 60-year-old man comes to the emergency department because of severe abdominal pain and nausea. The pain started 4 hours ago around the umbilical region and is now intensifies and shifted to the right iliac fossa. Physical examination shows positive psoas test and rebound tenderness at McBurney's point. Which of the following nerve fibers is responsible for mediating the pain from the inflamed organ, and what is their most likely pathway?

- A. Visceral afferents traveling with the vagus nerve
- B. Somatic efferents traveling with the T10 nerve
- C. Visceral afferents traveling via the superior mesenteric plexus
- D. Somatic afferents traveling in the T10 nerves
- E. Visceral efferents traveling in the superior mesenteric plexus

Describe the sympathetic, parasympathetic and sensory innervation of the foregut, midgut and hind gut.

Describe the location of the referred pain from the gastrointestinal tract and organs based on innervation and embryological origin.

A 22-year-old woman delivers a preterm baby girl at 30 weeks. Physical examination of the baby shows loops of small intestines enclosed in a membrane protruding through the anterior abdominal wall as seen in the image. Which of the following peritoneal spaces most likely contain these herniated loops of bowels?



- A. Omental bursa (lesser sac)**
- B. Greater sac**
- C. Hepatorenal recess (Morrison's pouch)**
- D. Paracolic gutters**
- E. Retrouterine pouch(Douglas)**

Discuss the embryonic processes that fail leading to development of an omphalocele. Describe the extent of the peritoneal cavity, the greater and lesser sacs, and the omental (epiploic) foramen (of Winslow).

A 75-year-old man comes to the emergency department because of nausea, vomiting, intense abdominal pain and inability to pass stool for the last two days. Patient history reveals unintentional weight loss of 25 pounds over the last two months. Physical examination shows pale mucous membranes and distended abdomen with hyperactive bowel sounds. On deep palpation of the abdomen there is rebound tenderness. Rectal examination shows no stool in the anal canal. Diagnostic studies show cancer of the colon extending from the left colic flexure to the middle of the descending colon. Colonic resection (surgery) is planned to remove the diseased portion of the colon. Which of the following arteries was most likely ligated during the surgery?

- A. Internal iliac**
- B. Celiac**
- C. Superior mesenteric**
- D. Inferior mesenteric**
- E. Common iliac**

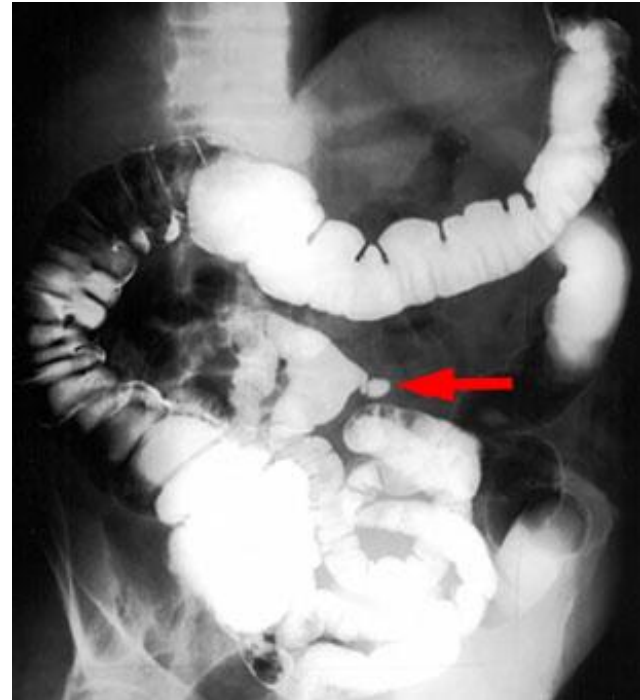
Describe the blood supply and the venous and lymphatic drainage of the large intestine.

A 60-year-old man comes to the emergency department because of severe abdominal pain and bloody stools for the past 2 days. History reveals chronic hypertension, hyperlipidemia and diabetes. His vitals on admission were: BP:160/90, RR16x1, PR:100x1. Physical examination shows a distended abdomen with absent bowel sounds. CT scan shows partial occlusion of the superior mesenteric artery and signs of colonic ischemia. Which part of the colon will most likely show the greatest amounts of ischemic changes?

- A. Cecum**
- B. Ascending colon**
- C. Right colic flexure**
- D. Left colic flexure**
- E. Transverse colon**

Describe the blood supply and the venous and lymphatic drainage of the large intestine.

A 25-year-old man comes to the emergency department because of severe paraumbilical pain and high fever. Physical examination of the abdomen reveal muscle guarding and rebound tenderness. Barium studies show an abnormal projection represented by the arrow in the image provided. Surgery is planned. A branch of which of the following arteries will most likely be ligated if this structure is to be removed?



- A. Celiac trunk
- B. Superior epigastric
- C. Inferior mesenteric
- D. Superior mesenteric
- E. Inferior epigastric

Describe the blood supply and the venous and lymphatic drainage of the duodenum, jejunum and ileum.

Describe the development of an ileal (Meckel) diverticulum.

A 67-year-old man came to the emergency department because of intense abdominal pain and blood in his vomit. History reveals chronic peptic ulcer disease and non-compliance with prescribed diet and medication. His vitals on admission were: BP:90/60 RR:16x1 PR:100x1, temperature:38.1°C (100.6°F). Physical examination shows pale mucous membranes and board like rigidity of the abdominal wall on superficial palpation. There is also rebound tenderness on deep palpation of the abdomen. Rectal examination shows “coffee ground” stool material on the gloved finger. CT scan shows perforation of the posterior wall of the stomach. In which of the following spaces will the stomach contents first accumulate?

- A. Greater sac**
- B. Lesser sac**
- C. Retroperitoneal**
- D. Pouch of Douglas**
- E. Splenic**

Describe the extent of the peritoneal cavity, the greater and lesser sacs, and the omental (epiploic) foramen (of Winslow).

A 56-year-old man comes to the physician because he has been feeling tired for the last 3 months. He complains that he has no appetite and has lost 20 pounds over the last month. Physical examination shows pale mucous membranes. He has a family history of colon cancer and a colonoscopy is performed. Diagnosis of right sided colon cancer is made and surgical resection and lymph node biopsy is performed. Which of the following lymph nodes will most likely be first to receive metastasized cells from the affected area?

- A. Lumbar
- B. Superior mesenteric
- C. Supraclavicular
- D. Inferior mesenteric
- E. Celiac

Describe the blood supply and the venous and lymphatic drainage of the large intestine.

Answer key

1. E
2. B
3. D
4. C
5. B
6. D
7. D
8. D
9. B
10. B